

Chapter 4

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1 Limits of Functions

Def 1. Let X and Y be metric spaces; suppose $E \subset X$, f maps E into Y , and p is a limit point of E . We write $f(x) \rightarrow q$ as $x \rightarrow p$, or

$$\lim_{x \rightarrow p} f(x) = q$$

if there is a point $q \in Y$ with the following property: For every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon$$

for all points $x \in E$ for which

$$0 < d_X(x, p) < \delta.$$

The symbols d_X and d_Y refer to the distances in X and Y , respectively.

Thm 2. Let X, Y, E, f , and p be as in the definition above. Then

$$\lim_{x \rightarrow p} f(x) = q$$

if and only if

$$\lim_{n \rightarrow \infty} f(p_n) = q$$

for every sequence (p_n) in E such that

$$p_n \neq p, \quad \lim_{n \rightarrow \infty} p_n = p.$$

Cor 3. If f has a limit at p , this limit is unique.

Def 4. Suppose we have two complex functions, f and g , both defined on E . By $f + g$ we mean the function which assigns to each point x of E the number $f(x) + g(x)$. Similarly we define the difference $f - g$, the product fg , and the quotient f/g of two functions, with the understanding that the quotient is defined only at those points x of E at which $g(x) \neq 0$. If f assigns to each point x of E the same number c , then f is said to be a constant function, or simply a constant, and we write $f = c$. If f and g are real functions, and if $f(x) \geq g(x)$ for every $x \in E$, we shall sometimes write $f \geq g$ for brevity.

Similarly, if $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, we define $\mathbf{f} + \mathbf{g}$ and $\mathbf{f} \cdot \mathbf{g}$ by

$$(\mathbf{f} + \mathbf{g})(x) = \mathbf{f}(x) + \mathbf{g}(x), \quad (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{f}(x) \cdot \mathbf{g}(x);$$

and if λ is a real number, $(\lambda \mathbf{f})(x) = \lambda \mathbf{f}(x)$.

Thm 5. Suppose $E \subset X$, a metric space, p is a limit point E , f and g are complex functions on E , and

$$\lim_{x \rightarrow p} f(x) = A, \quad \lim_{x \rightarrow p} g(x) = B.$$

Then

- (a) $\lim_{x \rightarrow p} (f + g)(x) = A + B$;
- (b) $\lim_{x \rightarrow p} (fg)(x) = AB$;
- (c) $\lim_{x \rightarrow p} (f/g)(x) = \frac{A}{B}$, if $B \neq 0$.

Rmk 6. If $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, then (a) remains true, and (b) becomes

$$(b') \lim_{x \rightarrow p} (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{A} \cdot \mathbf{B}.$$

2 Continuous Functions

Def 7. Suppose X and Y are metric spaces, $E \subset X$, $p \in E$, and f maps E into Y . Then f is said to be *continuous at p* if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \varepsilon$$

for all points $x \in E$ for which $d_X(x, p) < \delta$.

If f is continuous at every point of E , then f is said to be *continuous on E* .

Thm 8. In the situation given above, assume also that p is a limit point of E . Then f is continuous at p if and only if $\lim_{x \rightarrow p} f(x) = f(p)$

Thm 9. Suppose X, Y, Z are metric spaces, $E \subset X$, f maps E into Y , g maps the range of f , $f(E)$, into Z , and h is the mapping of E into Z defined by

$$h(x) = g(f(x)) \quad (x \in E).$$

If f is continuous at a point $p \in E$ and if g is continuous at the point $f(p)$, then h is continuous at p .

This function h is called the *composition* or the *composite* of f and g . The notation

$$h = g \circ f$$

is frequently used in this context.

Thm 10. A mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y .

Cor 11. A mapping f of a metric space X into a metric space Y is continuous if and only if $f^{-1}(C)$ is closed in X for every closed set C in Y .

Thm 12. Let f and g be complex continuous functions on a metric space X . Then $f + g$, fg , and f/g are continuous on X .

Thm 13.

(a) Let $f_1, \dots, f_k$ be real functions on a metric space X , and let $\mathbf{f}$ be the mapping of X into $\mathbb{R}^k$ defined by

$$\mathbf{f}(x) = (f_1(x), \dots, f_k(x)) \quad (x \in X);$$

then $\mathbf{f}$ is continuous if and only if each of the functions $f_1, \dots, f_k$ is continuous.

(b) If $\mathbf{f}$ and $\mathbf{g}$ are continuous mappings of X into $\mathbb{R}^k$, then $\mathbf{f} + \mathbf{g}$ and $\mathbf{f} \cdot \mathbf{g}$ are continuous on X .

The functions $f_1, \dots, f_k$ are called the *components* of $\mathbf{f}$. Note that $\mathbf{f} + \mathbf{g}$ is a mapping into $\mathbb{R}^k$, whereas $\mathbf{f} \cdot \mathbf{g}$ is a *real* function on X .

3 Continuity and Compactness

Def 14. A mapping $\mathbf{f}$ of a set E into $\mathbb{R}^k$ is said to be *bounded* if there is a real number M such that $|\mathbf{f}(x)| \leq M$ for all $x \in E$.

Thm 15. Suppose f is a continuous mapping of a compact metric space X into a metric space Y . Then $f(X)$ is compact.

Thm 16. If f is a continuous mapping of a compact metric space X into $\mathbb{R}^k$, then $\mathbf{f}(x)$ is closed and bounded. Thus, $\mathbf{f}$ is bounded.

Thm 17. Suppose f is a continuous real function on a compact metric space X , and

$$M = \sup_{p \in X} f(p), \quad m = \inf_{p \in X} f(p).$$

Then there exist points $p, q \in X$ such that $f(p) = M$ and $f(q) = m$.

Thm 18. Suppose f is a continuous 1-1 mapping of a compact metric space X onto a metric space Y . Then the inverse mapping f^{-1} defined on Y by

$$f^{-1}(f(x)) = x \quad (x \in X)$$

is a continuous mapping of Y onto X .

Def 19. Let f be a mapping of a metric space X into a metric space Y . We say that f is *uniformly continuous* on X if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_Y(f(p), f(q)) < \varepsilon$$

for all p and q in X for which $d_X(p, q) < \delta$.

Thm 20. Let f be a continuous mapping of a compact metric space X into a metric space Y . Then f is uniformly continuous on X .

Thm 21. Let E be a noncompact set in $\mathbb{R}^1$. Then

- (a) there exists a continuous function on E which is not bounded.
- (b) there exists a continuous and bounded function on E which has no maximum.

If, in addition, E is bounded, then

- (c) there exists a continuous function on E which is not uniformly continuous.

4 Continuity and Connectedness

Thm 22. If f is a continuous mapping of a metric space X into a metric space Y , and if E is a connected subset of X , then $f(E)$ is connected.

Thm 23. Let f be a continuous real function on the interval $[a, b]$. If $f(a) < f(b)$ and if c is a number such that $f(a) < c < f(b)$, then there exists a point $x \in (a, b)$ such that $f(x) = c$.

5 Discontinuities

Def 24. Let f be defined on (a, b) . Consider any point x such that $a \leq x < b$. We write

$$f(x+) = q$$

if $f(t_n) \rightarrow q$ as $n \rightarrow \infty$, for all sequences (t_n) in (x, b) such that $t_n \rightarrow x$. To obtain the definition of $f(x-)$, for $a < x \leq b$, we restrict ourselves to sequences (t_n) in (a, x) .

It is clear that for any point x of (a, b) , $\lim_{t \rightarrow x} f(t)$ exists if and only if

$$f(x+) = f(x-) = \lim_{t \rightarrow x} f(t).$$

Def 25. Let f be defined on (a, b) . If f is discontinuous at a point x , and if $f(x+)$ and $f(x-)$ exist, then f is said to have a discontinuity of the *first kind*, or a *simple discontinuity*, at x . Otherwise the discontinuity is said to be of the *second kind*.

6 Monotonic Functions

Def 26. Let f be real on (a, b) . Then f is said to be *monotonically increasing* on (a, b) if $a < x < y < b$ implies $f(x) \leq f(y)$. If the last inequality is reversed, we obtain the definition of a *monotonically decreasing* function.

Thm 27. Let f be monotonically increasing on (a, b) . Then $f(x+)$ and $f(x-)$ exist at every point of x of (a, b) . More precisely,

$$\sup_{a < t < x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x < t < b} f(t).$$

Rmk 28. Given any countable subset E of (a, b) , which may even be dense, we can construct a function f , monotonic on (a, b) , discontinuous at every point of E , and at no other point of (a, b) .

Def 29. Given a function f on (a, b) ,

- (a) If $f(x-) = f(x)$ at all points of (a, b) , then f is *continuous from the left*.
- (b) If $f(x+) = f(x)$ at all points of (a, b) , then f is *continuous from the right*.

7 Infinite Limits and Limits at Infinity

Def 30. For any real c , the set of real numbers x such that $x > c$ is called a neighborhood of $+\infty$ and is written $(c, +\infty)$. Similarly, the set $(-\infty, c)$ is a neighborhood of $-\infty$.

Def 31. Let f be a real function defined on $E \subset \mathbb{R}$. We say that

$$f(t) \rightarrow A \text{ as } t \rightarrow x,$$

where A and x are in the extended real number system, if for every neighborhood U of A there is a neighborhood V of x such that $V \cap E$ is not empty, and such that $f(t) \in U$ for all $t \in V \cap E$, $t \neq x$.

Thm 32. Let f and g be defined on $E \subset \mathbb{R}$. Suppose

$$f(t) \rightarrow A, \quad g(t) \rightarrow B \quad \text{as } t \rightarrow x$$

Then

- (a) $f(t) \rightarrow A'$ implies $A' = A$,
- (b) $(f + g)(t) \rightarrow A + B$,
- (c) $(fg)(t) \rightarrow AB$,
- (d) $(f/g)(t) \rightarrow A/B$,

provided the right members of (b), (c) and (d) are defined.

8 Exercise Notes

Thm 33. A continuous function is determined by its values on a dense subset of its domain.

Def 34. If f is a real continuous function defined on a closed set $E \subset \mathbb{R}^1$, and there exists a continuous real function g on $\mathbb{R}^1$ such that $g(x) = f(x)$ for all $x \in E$, then g is called *continuous extensions* of f from E to $\mathbb{R}^1$.

Def 35. If f is a real continuous function defined on X , and there exists a continuous function g on $E \subset X$ such that $g(x) = f(x)$ for all $x \in E$, then g is called the *restriction* of f .

Def 36. If E is a nonempty subset of a metric space X , define the distance from $x \in X$ to E by

$$\rho_E(x) = \inf_{z \in E} d(x, z)$$

Thm 37.

$$|\rho_E(x) - \rho_E(y)| \leq d(x, y)$$

for all $x \in X, y \in X$.

Thm 38. Suppose K and F are disjoint sets in a metric space X , K is compact, F is closed. Then there exists $\delta > 0$ such that $d(p, q) > \delta$ if $p \in K, q \in F$.

Def 39. The property that pairs of disjoint closed sets in metric spaces can be covered by pairs of disjoint open sets is called *normality*.

Def 40. A real-valued function f defined in (a, b) is said to be *convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

whenever $a < x < b, a < y < b$, and $0 < \lambda < 1$.

Thm 41. Every convex function is continuous.

Thm 42. Let α be an irrational real number. Let C_1 be the set of all integers, let C_2 be the set of all $n\alpha$ with $n \in C_1$. Then $C_1 + C_2$ is a countable dense subset of $\mathbb{R}^1$.